Experiment 2 – Basic Networking Operations and Troubleshooting

Learning Objective: Students should be able to use basic networking commands in Linux (ifconfig, tracert, nslookup, netstat, ARP, IP, dig, route)

Tools: Ubuntu, MS Word

Theory:

Linux networking commands are essential tools used to configure, monitor, and troubleshoot network settings on a system. They help manage network interfaces, test connectivity, analyze traffic, and perform DNS lookups. Commands like ping, ifconfig, ip, netstat, and traceroute allow users to check network connections, view interface details, and verify routes. Advanced tools such as dig and ss provide deeper insights into DNS records, active sockets, and port usage. These commands are widely used by system administrators for configuring and debugging Linux networks effectively.

They play a crucial role in diagnosing network issues, managing IP addresses, and ensuring smooth communication between devices on a network. Commands like hostname and uptime also provide valuable system-related network information. Mastering these commands enables users to efficiently manage Linux-based networks and servers, making them indispensable for network management and troubleshooting tasks.

Commands:

1. **ip:** Displays and manipulates network interfaces, routing, and tunnels.
   Example: ip a - Shows all network interfaces and their IPs.

2. **ip addr:** Shows IP address details of all or specific network interfaces.
   Example: ip addr show - Lists IPs assigned to interfaces

3. **ip link:** Displays and manages network interface status.
   Example: ip link set eth0 down - Disables the eth0 interface.

4. **ip route:** Displays or modifies the system's routing table.
   Example: ip route show - Lists current routing table entries.

5. **dig:** Fetches DNS records to test name resolution.
   Example: dig google.com - Retrieves DNS information for google.com.

6. **host**: Performs DNS lookups to resolve hostnames and IPs.
   Example: host 8.8.8.8 - Finds the hostname for IP 8.8.8.8.

7. **hostname:** Displays or sets the system's hostname.
   Example: hostname - Shows the current hostname.

8. **ss:** Displays socket statistics and shows active connections.
   Example: ss -tuln - Lists all listening TCP/UDP ports.

9. **whoami:** Prints the username of the current logged-in user.
   Example: whoami - Shows the current user's name.

10. **mtr:** Combines traceroute and ping to analyze network connectivity.
    Example: mtr google.com - Traces route and monitors packet loss to google.com.

11. **date:** Displays or sets the system's date and time.
    Example: date - Shows the current date and time.

12. **history:** Shows the list of previously executed commands in the terminal.
    Example: history - Displays the command history.

13. **uname -r:** Displays the kernel version of the operating system.
    Example: uname -r - Shows the current Linux kernel version.

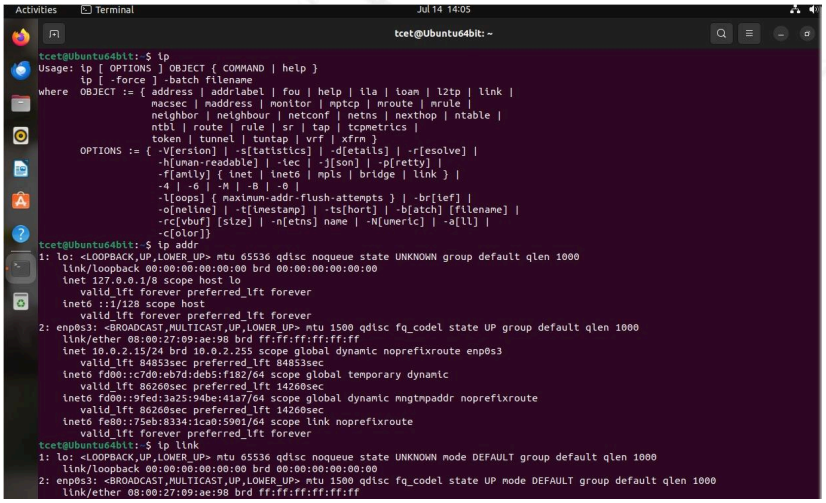
14. **uptime:** Shows how long the system has been running and load averages.
    Example: uptime - Displays system uptime and load info.

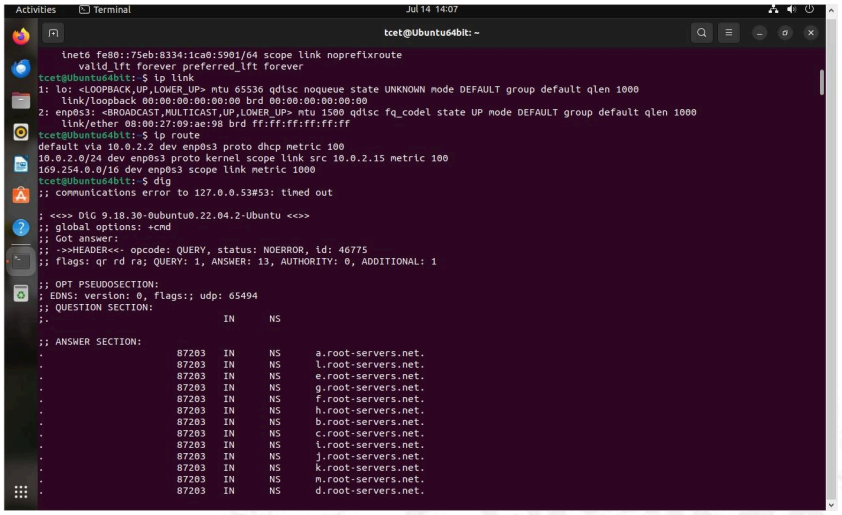
15. **echo "Hello":** Prints text or variables to the terminal.
    Example: echo "hello" - Prints the word "hello" on the screen.

Result and Discussion:

Write the output of commands ip, ip addr, ip link, ip route, dig, host, hostname, ss, whoami, mtr, date, history, uname -r, uptime, and echo "Hello" in the following syntax

Output:





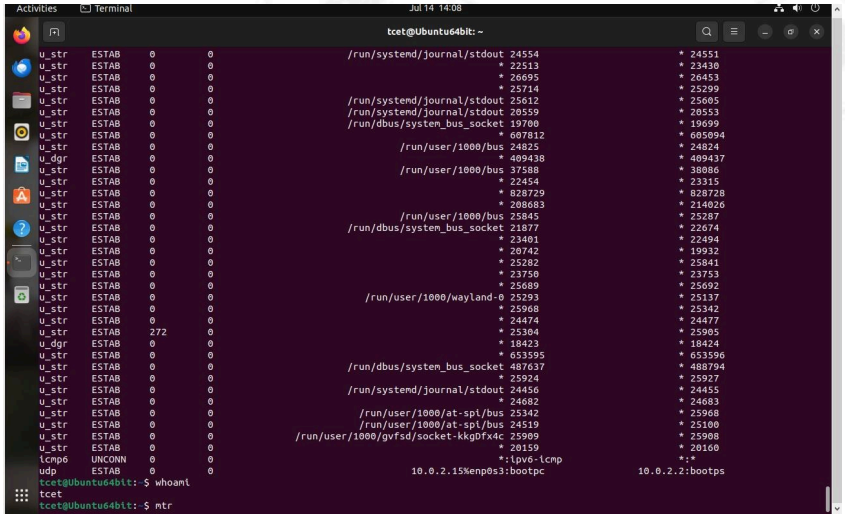

```
    inet6 fe80::75eb:8334:1ca0:5901/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
tcet@Ubuntu64bit:~$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:09:ae:98 brd ff:ff:ff:ff:ff:ff
tcet@Ubuntu64bit:~$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
169.254.0.0/16 dev enp0s3 scope link metric 1000
tcet@Ubuntu64bit:~$ dig
;; communications error to 127.0.0.53#53: timed out

; <<>> DiG 9.18.30-0ubuntu0.22.04.2-Ubuntu <<>>
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 46775
;; flags: qr rd ra; QUERY: 1, ANSWER: 13, AUTHORITY: 0, ADDITIONAL: 1

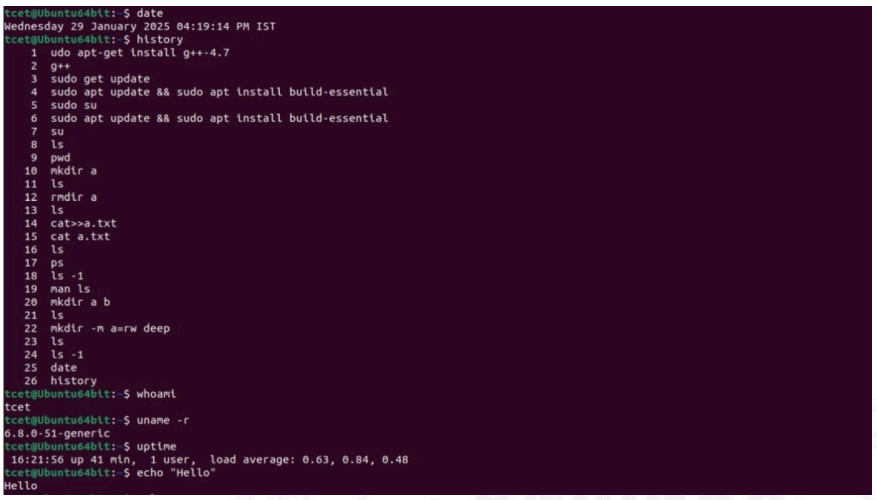
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;.                              IN      NS

;; ANSWER SECTION:
.                       87203   IN      NS      a.root-servers.net.
.                       87203   IN      NS      l.root-servers.net.
.                       87203   IN      NS      e.root-servers.net.
.                       87203   IN      NS      g.root-servers.net.
.                       87203   IN      NS      f.root-servers.net.
.                       87203   IN      NS      h.root-servers.net.
.                       87203   IN      NS      b.root-servers.net.
.                       87203   IN      NS      c.root-servers.net.
.                       87203   IN      NS      i.root-servers.net.
.                       87203   IN      NS      j.root-servers.net.
.                       87203   IN      NS      k.root-servers.net.
.                       87203   IN      NS      m.root-servers.net.
.                       87203   IN      NS      d.root-servers.net.
```

```
tcet@Ubuntu64bit:~$ host
Usage: host [-aCdlrTvVw] [-c class] [-N ndots] [-t type] [-W time]
            [-R number] [-m flag] [-p port] hostname [server]
       -a is equivalent to -v -t ANY
       -A is like -a but omits RRSIG, NSEC, NSEC3
       -c specifies query class for non-IN data
       -C compares SOA records on authoritative nameservers
       -d is equivalent to -v
       -l lists all hosts in a domain, using AXFR
       -m set memory debugging flag (trace|record|usage)
       -N changes the number of dots allowed before root lookup is done
       -p specifies the port on the server to query
       -r disables recursive processing
       -R specifies number of retries for UDP packets
       -s a SERVFAIL response should stop query
       -t specifies the query type
       -T enables TCP/IP mode
       -U enables UDP mode
       -v enables verbose output
       -V print version number and exit
       -w specifies to wait forever for a reply
       -W specifies how long to wait for a reply
       -4 use IPv4 query transport only
       -6 use IPv6 query transport only
tcet@Ubuntu64bit:~$ hostname
Ubuntu64bit
tcet@Ubuntu64bit:~$ ss
Netid     State      Recv-Q     Send-Q                                  Local Address:Port                 Peer Address:Port       Process
u_str     ESTAB      0          0                                                   * 1023154                         * 1023155
u_str     ESTAB      0          0                                                   * 409539                          * 409540
u_str     ESTAB      0          0                                                   * 44569                           * 0
u_str     ESTAB      0          0                                                   * 26210                           * 26293
u_str     ESTAB      0          0                                                   * 25971                           * 25974
u_str     ESTAB      0          0                                                   * 19888                           * 19889
u_dgr     ESTAB      0          0                                                   * 25225                           * 15220
u_str     ESTAB      0          0                                                   * 25181                           * 25190
u_str     ESTAB      0          0                          /run/systemd/journal/stdout 20659                         * 19770
u_str     ESTAB      0          0                          /run/systemd/journal/stdout 18427                         * 19439
u_str     ESTAB      0          0                                                   * 409916                          * 409285
```




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u_str     ESTAB      0          0                          /run/systemd/journal/stdout 24554                         * 24551
u_str     ESTAB      0          0                                                   * 22513                           * 23430
u_str     ESTAB      0          0                                                   * 26695                           * 26453
u_str     ESTAB      0          0                                                   * 25714                           * 25299
u_str     ESTAB      0          0                          /run/systemd/journal/stdout 25612                         * 25605
u_str     ESTAB      0          0                          /run/systemd/journal/stdout 20559                         * 20553
u_str     ESTAB      0          0                          /run/dbus/system_bus_socket 19700                         * 19699
u_str     ESTAB      0          0                                                   * 607812                          * 605094
u_str     ESTAB      0          0                                  /run/user/1000/bus 24825                         * 24824
u_dgr     ESTAB      0          0                                                   * 409438                          * 409437
u_str     ESTAB      0          0                                  /run/user/1000/bus 37588                         * 38086
u_str     ESTAB      0          0                                                   * 22454                           * 23315
u_str     ESTAB      0          0                                                   * 828729                          * 828728
u_str     ESTAB      0          0                                                   * 208683                          * 214026
u_str     ESTAB      0          0                                  /run/user/1000/bus 25845                         * 25287
u_str     ESTAB      0          0                          /run/dbus/system_bus_socket 21877                         * 22674
u_str     ESTAB      0          0                                                   * 23401                           * 22494
u_str     ESTAB      0          0                                                   * 20742                           * 19932
u_str     ESTAB      0          0                                                   * 25282                           * 25841
u_str     ESTAB      0          0                                                   * 23750                           * 23753
u_str     ESTAB      0          0                                                   * 25689                           * 25692
u_str     ESTAB      0          0                            /run/user/1000/wayland-0 25293                         * 25137
u_str     ESTAB      0          0                                                   * 25968                           * 25342
u_str     ESTAB      0          0                                                   * 24474                           * 24477
u_str     ESTAB      272        0                                                   * 25304                           * 25905
u_dgr     ESTAB      0          0                                                   * 18423                           * 18424
u_str     ESTAB      0          0                                                   * 653595                          * 653596
u_str     ESTAB      0          0                          /run/dbus/system_bus_socket 487637                        * 488794
u_str     ESTAB      0          0                                                   * 25924                           * 25927
u_str     ESTAB      0          0                          /run/systemd/journal/stdout 24456                         * 24455
u_str     ESTAB      0          0                                                   * 24682                           * 24683
u_str     ESTAB      0          0                            /run/user/1000/at-spi/bus 25342                         * 25968
u_str     ESTAB      0          0                            /run/user/1000/at-spi/bus 24519                         * 25100
u_str     ESTAB      0          0                  /run/user/1000/gvfsd/socket-kkgDfx4c 25909                         * 25908
u_str     ESTAB      0          0                                                   * 20159                           * 20160
icmp6     UNCONN     0          0                                                   *:ipv6-icmp                       *:*
udp       ESTAB      0          0                                        10.0.2.15%enp0s3:bootpc                10.0.2.2:bootps
tcet@Ubuntu64bit:~$ whoami
tcet
tcet@Ubuntu64bit:~$ mtr
```



Learning Outcomes: The student should have the ability to:

LO4.1 Outline the basic networking commands.
LO4.2 Express the functions of every command.
LO4.3 Illustrate the basic networking operations and troubleshooting.

Course Outcomes: Upon completion of the course, students will be able to use and understand basic networking commands in Linux.

Conclusion:

Viva Questions:

1. Why is ping used?
2. Difference between ARP and RARP.
3. Why is tracert used?
4. What are the five columns in tracert?

For Faculty Use:

Correction Parameters	Formative Assessment [40%]	Timely completion of Practical [40%]	Attendance / Learning Attitude [20%]	
Marks Obtained				

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